

UAV TARGET TRACKING BY DETECTION VIA DEEP NEURAL NETWORKS

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ABSTRACT

This paper introduces an Image Based Visual Servoing (IBVS) system to track a ground target from a flying UAV. We implement a tracking system via two deep neural networks; approaching network and chasing network. In the approaching network, the UAV-target relative distance is estimated through the tracked target state to drive the UAV to a desired distance. In the chasing network, the tracked target orientation is used in resolving the ambiguity between the UAV's yawing and lateral movements and aligning the UAV with the tracked target heading to ease the tracking task. The detection accuracy for both networks has been validated and a series of real flight data, which has been conducted on ground targets and persons showed the proposed system efficiency in target tracking and following.

Index Terms— IBVS, UAV-target following, two deep networks, approaching and chasing.

1. INTRODUCTION

The UAVs are equipped with a variety of sensors to accomplish various missions in civil and military applications. Particularly, the camera as a cheap passive sensor with a negligible payload and its robustness against jamming compared with the GPS leads to utilizing it in numerous UAV tasks e.g., visual navigation [1], stabilization [2], and servoing [3].

In the vision-based servoing task, the UAV utilizes the tracked visual features to control its motion. Autonomously piloting the UAV based on the visual features has been proposed recently in vast research works [4]. Researchers exploit the new findings in the computer vision field to grant the vision-based controlling system higher efficiency.

Recently, deep learning has been contributed for solving many computer vision tasks. This contribution led to an extraordinary enhancement in vision-based system performance. The motivation of this work is designing and implementing a full vision-based target tracking and chasing system while utilizing the deep detection network as a backbone for this task. Our contribution can be summarized as follows:

- To the extent of our knowledge, this is the first vision-based proposal divides the tracking and following task into two successive modes based on two mutual deep networks to realize how to approach the target before chasing it and resolve the ambiguity of UAV's yawing-lateral movement.
- The proposed system successfully integrates the detection network with the particle filter as a tracking-by-detection paradigm and exploits the tracked feature for servoing the parrot bebop 2 to follow a target in real time.
- A ground target dataset utilizing three prototypes (Jeep, Tank, and Aircraft) was built to facilitate the training of both networks and will be publicly online to aid the research work.

2. RELATED WORKS

A notable amount of research works analyzed the target tracking and following task. Mainly, they addressed how to strengthen the visual tracker to obtain appropriate visual features for the servoing task. The conventional template matching was used by [5] to achieve the target localization. Then, the density-based color histograms descriptors are used in [6] to grant the tracker more robust target representations.

Although, a full vision-based controlling system was presented by [7] but it used the conventional binary image segmentation to localize the desired target. Moreover, the whole system was built in simulator and has not been examined in reality. A decent IBVS system [8] was presented for the target following, however the assumption of the UAV-target fixed distance limits its performance in the following task.

3. THE PROPOSED SYSTEM

As illustrated in Figure 1, the proposed system works in two networks, i.e., approaching network and chasing network. Each network is a merely IBVS proposal for controlling the UAV according to the visual desired target's features. These features are built from the detection results of the deep neural network in each phase then tracked (predicted and updated) through the particle filter framework. Both approaching and chasing controllers drive the UAV

according to the error between the reference and the tracked features to accomplish the following task. The tiny-YOLO [9] detection framework is used for both networks due to its compromise between the reasonable accuracy and the inference time and all system components were implemented in the Robot Operating System (ROS) framework as nodes.

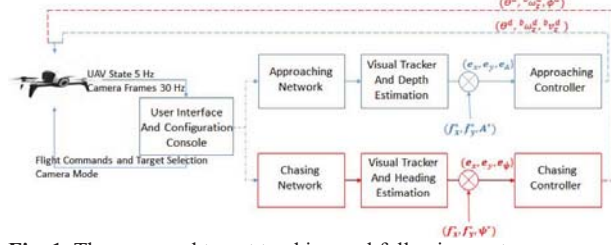


Fig. 1. The proposed target tracking and following system.

3.1. Depth estimation

Equivalent to the supposed methodology in [10], the relative UAV-target distance is estimated by the depicted trigonometric in Figure 2 with some given priors as follow:

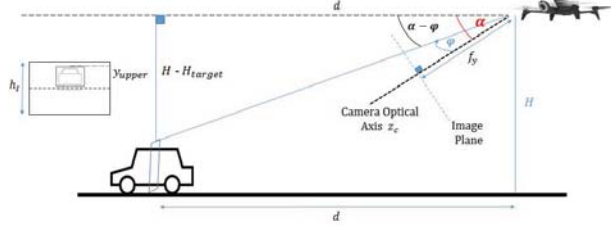


Fig. 2. The relative UAV-target distance estimation.

- H_{target} which is the height of the ground target in real life measured in meter, and there is a lookup table for inquiring the target height according to the class label of the detection result.
- H_{UAV} which is measured by the UAV's barometric sensor and acquired from the UAV state.
- α is the current UAV's camera tilt angle which is measured between the UAV's X-axis inertial frame and its camera optical axis.
- The UAV vertical Field of View (FOV) β which is determined from the camera intrinsic parameters after its calibration and is given by

$$\beta = 2 \tan^{-1}\left(\frac{h_l}{2f_y}\right) \quad (1)$$

where h_l is the image height in pixels, and f_y is the camera focal length in camera y-axis direction. Under the assumption that, the bounding box holds the whole target up to the ground and perpendicular to it. Moreover, the ground itself is flat (horizontal), the depth of the ground target can be determined via the simplified trigonometric shown in Figure 2 as

$$d = \frac{H - H_{target}}{\tan(\alpha - \varphi)} \quad (2)$$

where φ is the angle between the projected beam of the detected target's bounding box upper edge y_{upper} and the UAV's camera optical axis, and it is given by

$$\varphi = \frac{(h_l/2 - y_{upper})}{h_l} \beta \quad (3)$$

3.2. Approaching and chasing networks

The main task of the detector is to localize and classify the desired target in the input image. To successfully detect the target in any application system, the detector should tackle the challenges imposed by the environment of that application. These challenges could be a wide variation in the target appearance, target out-of-plane rotation or even more the target is very small to be detected. In this regard, the challenges in approaching and chasing networks are considered as follow:

3.2.1. Approaching network

In the approaching network, the UAV is searching for a ground target with its eye (camera) is looking forward. This mode requires a detector has the ability to early detect the target i.e. detection of more far (small) target is better. Moreover, as a result of the looking forward camera (approximately, 13 degrees off the UAV x-axis body frame), this attitude viewpoint gives the target different appearance, it may consider as an out-of-plane rotation of the ground target.

The approaching network also should pay attention to the unknown target attitude, which reflects the requirement of detecting the target from any approaching point i.e. considering the target is placed at a circle center with any heading angle and the UAV at any point on that circle circumference has the ability to detect it.

To that end, a particular dataset for three prototypes ground targets (jeep, tank, and aircraft) was built for the sake of comprehensive network training. The dataset covers various targets attitudes, sizes and perspective angles to meet the requirements of this mode for early target detection from any arbitrary approaching point.

3.2.2. Chasing network

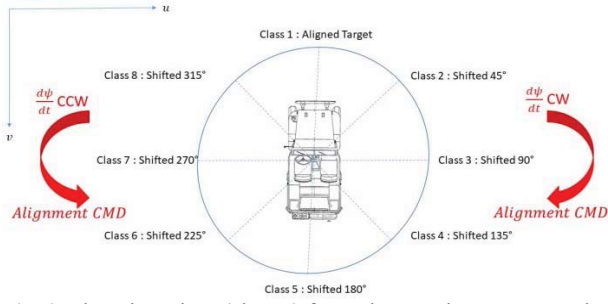


Fig. 3. The orientations (classes) for each ground target are used to facilitate the UAV-target alignment in the chasing network.

After reaching the target, the UAV is switched to the chasing network automatically. In that mode, the camera pan tilt module will rotate the camera automatically in the vertical plane to a "looking down position" semi-vertical attitude viewpoint. Presently, the UAV rides the target and waits for the target to move to chase it. For the sake of ease the chasing task for the IBVS controller, the presented Deep-patch Orientation Network (DON) [11] was utilized in the chasing network. Thus, at the beginning of that mode the UAV will know the target relative orientation with respect to it, then it will yaw to align its camera y-axis y_c " which is almost parallel to the UAV x-axis x_b " to the target heading attitude as shown in Figure 3. Subsequently, at the moment of the "aligned" locked on target movement, the UAV will track and follow it directly.

To achieve the aforementioned behavior, more than 24 videos sequences were utilized in training of the DON chasing network [12]. All of them examine the possible orientations of the three prototypes targets from their plane view as demonstrated in the Figure 3. Some samples from the training set of both networks are shown in Figure 4.

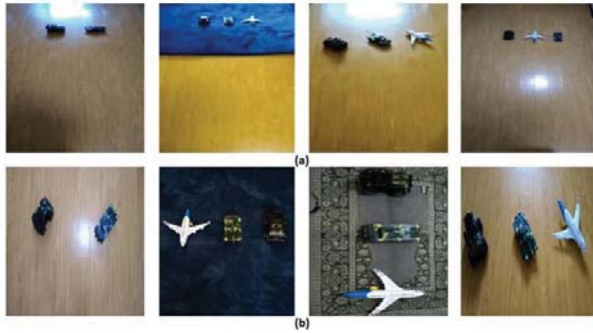


Fig. 4. Samples of three ground targets prototypes from the training set of both networks in (a) approaching network and (b) chasing network.

3.3. Target state modeling and tracking process

The proposed tracking system works inside the boundaries of the tracking-by-detection paradigm for a single target

tracking. The (measurements) detection results are delivered to an appropriate particle filter which is robust in case of motion blur due to abrupt vibration in the UAV motion or transmission signal errors between the UAV and the ground station.

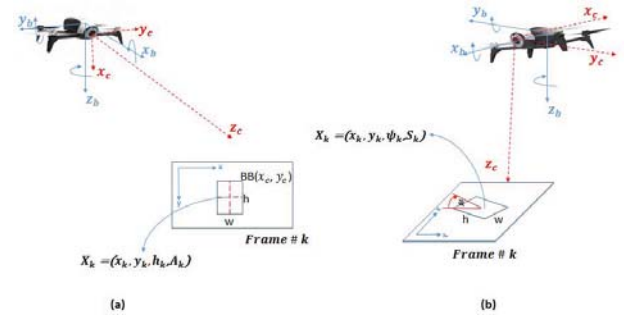


Fig. 5. Target state modeling in each network. (a) In the approaching network "with a forward-looking camera" the target height is tracked due to its importance in estimating the depth. (b) In the chasing network "with a down-looking camera", the target heading is tracked to facilitate the servoing task and resolve the ambiguity.

As depicted in Figure 5 (a), the UAV in the approaching network "with a forward-looking camera" should exploit the tracking results to reach the target. Thus, the target state should be picked to facilitate the achievement of that goal. Therefore, the target state at frame number k is defined as

$$x_k = [x_k, y_k, h_k, A_k] \quad (4)$$

where x_k and y_k are the target's bounding box center, h_k is the target bounding box height, and the A_k is the target area at frame number k . Note that, in this mode the target aspect ratio is assumed to be constant. The target height is tracked due to its importance in estimating the depth as referred in Section 3.1. Later, both target centers with its area are used as features to accomplish the vision-based controlling task which will be highlighted in the next section.

On the other hand, the target state in the chasing network is defined as

$$x_k = [x_k, y_k, \psi_k, S_k] \quad (5)$$

where ψ_k is the target heading angle which is delivered from the DON as a quantized direction then smoothed "tracked" in the upcoming frames by the tracker with a constant angular velocity motion model [12]. The target heading is tracked to facilitate the servoing task in that mode. And the target's aspect ratio S_k is tracked since it is correlated with the target orientation i.e. it will be changed as the target orientation altered.

All particles are initialized with equal weights ($1/N$) with a uniform distribution covering the whole scene and propagated according to a constant velocity model. In case of the detector failure, the "update" model will not be

gainful and the filter can stay predict the missed target state for a while based on the dynamic model only. In the proposed tracking process, if the tracker fails in the chasing network, the approaching network will be in charge of re-detect the missed target as it has more comprehensive appearance models for the ground targets.

3.4. Approaching and chasing vision based controller

In this section, both controllers are broken down to demonstrate how to achieve the required behavior. The proposed controllers are completely vision-based; therefore all measurements, feedbacks, and references signals are determined in the image plane as shown in Figure 6.

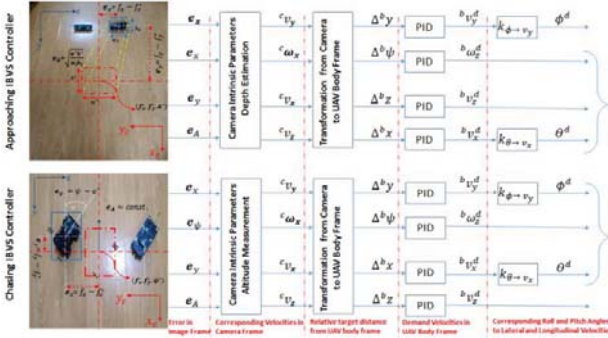


Fig. 6. The breakdown of the proposed IBVS controller in each network. Each channel PID controller is implemented in each network and is tuned online through an implemented GUI to set proper gain values for the controlled vision-based process.

The visual features are picked from the tracked target state x_k which is delivered by the tracker for each frame. The error signals (e_x, e_y, e_A, e_ψ) are the differences between the selected references ($f_x^*, f_y^*, f_A^*, \psi$) and the current tracked target features (f_x, f_y, f_A, ψ). The controller concerns only about the error signals and attempts to eliminate them via sending the desired command set of velocities ($b^d_{v_x}, b^d_{v_y}, b^d_{v_z}, b^d_{\omega_z}$) to the UAV's flight controller.

The UAV is an under-actuated system; it implies that some of its degrees of freedom cannot be controlled directly. Therefore, the longitudinal and lateral velocities are obtained by setting the corresponding pitch and roll angles, respectively. Thus, the controller desired command of velocities is sent as $(\theta^d, \phi^d, b^d_{v_z}, b^d_{\omega_z})$.

Each channel PID controller is implemented in each mode to set proper gain values for the controlled vision-based process. The PID tuning process can be set "online" within the proposed GUI which provides an appropriate way to establish the required system response.

The approaching controller "which is inspired by [13]" acts to drive the UAV from any position surrounds the ground target to a predefined distance with it. This distance is proportional to the desired target area in the image plane [14] i.e. the closer target, the more area it has and vice-versa. Three features (f_x, f_y, f_A) are established from the tracked target state which provides three independent constraints to the flight controller similar to the proposed features in [8].

The (f_x, f_y) feature points are the normalized target center which is determined from its tracked bounding box BB as

$$f_x = \frac{x_c}{w_I}, f_y = \frac{y_c}{h_I} \quad (6)$$

with the tracked target normalized area,

$$f_A = \sqrt{\frac{w_{BB} h_{BB}}{w_I h_I}} \quad (7)$$

where (x_c, y_c) is the BB center of the tracked target, with width w_{BB} and height h_{BB} . w_I and h_I are the width and the height of the input image, respectively. All of the aforementioned parameters are measured in pixels.

The "Ambiguity" in the vision-based controlling system is defined as the movement which causes the same effect in the image plane. In the approaching network, both yawing and lateral velocity will cause the same target movement in the image plane. As a result of this "Ambiguity", the lateral velocity command is discarded and only the yawing command is exploited in this mode.

On the other hand, the chasing controller acts to follow the ground target as it moves from the UAV's center field of view. During the chasing control, the virtual camera is panned to look downward (approximately 80 degrees). The depth is kept constant and is inferred from the UAV's barometric sensor. Three features are established from the tracked target state which provides three independent constraints to the flight controller to successfully follow the moving target.

The normalized center of the tracked target is utilized to provide the first and second feature points as it was taken in the approaching network. The third feature is the tracked target orientation which provides a constraint for the UAV yawing command to align with the ground target heading. It is worthy of note that, the DON's orientation information is utilized to remove the ambiguity between the lateral and yawing movement in this mode.

4. EXPERIMENTAL VALIDATION

For judging the overall system performance, the experiments are evaluated in two stages, one in terms of the visual detection system accuracy and the other regarding the proposed servoing behavior.

4.1. Visual detection accuracy

In the tracking-by-detection paradigm, especially with a motion-based tracker, the detector failure is almost a tracking system breakdown. Because of that, both deep networks are evaluated in terms of their detection accuracy. AS shown Figure 7, the evaluation is carried out as stated by the PASCAL VOC for target detection evaluation protocol. The approaching dataset was split into 3960, 2722 and 1483 images for training, validation, and testing, respectively and for the chasing network; it was 3159, 2171, and 1185, respectively.

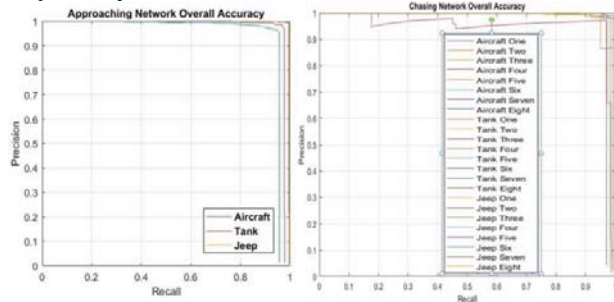


Fig. 7. The Precision-Recall curves for the three ground targets of the approaching network (left) and for each ground target's orientation of the chasing network (right).

The last convolutional layer feature map output was set 45 in the Tiny-YOLO configuration file which corresponding to three ground targets (Aircraft, Tank, and Jeep). On the other hand, for 24 classes (eight orientations for each class) it was set to 145 for the chasing network. The input resolution is 416 x 416 for both networks and the training was carried out for 45,000 iterations. The multi-scale training was activated to strengthen the ability of both networks to detect the ground targets in far/high and close/low depth values.

Table 1 summarizes the overall mean Average Precision (mAP) of the chasing network for each ground target's orientation, as mentioned in Figure 3, while Table 2 summarizes the accuracy of the approaching network for each ground target with different GT overlapping percentage, to assure its localization accuracy.

Table 1. The overall accuracy of the chasing network for each ground target's orientation (O), where T is Target and C is Class.

O T	C1	C2	C3	C4	C5	C6	C7	C8
Aircra ft	100%	99%	90%	91%	100 %	100 %	89%	90%
Tank	91%	90%	87%	91%	90%	100 %	80%	91%
Jeep	90%	100 %	77%	90%	70%	91%	76%	91%

Table 2. The overall accuracy of the approaching network with different GT overlapping percentage, where IoU is Intersection over Union

IoU%	50%	60%	70%	80%
T				
Aircraft	91%	91%	91%	78%
Tank	91%	91%	90%	77%
Jeep	91%	91%	90%	75%

4.2. Visual Servoing Accuracy

Real flight experiments were carried out to validate the required behavior in each network. In the approaching network, a ground target is approximately placed in a center of a depicted circle on a horizontal ground; the radius of this circle represents the desired UAV-target distance. The bebop 2.0 UAV is launched from some arbitrary points with different longitudinal and lateral displacements from that circle. Thus, it will fly from one of these points at each test time to reach the target. This scenario is established to approve the behavior of the proposed approaching network. Verifying the UAV ability to detect and approach the ground target from arbitrary horizontal points of view surrounding it. Figure 8 shows a 2D plot of the flight trajectories from arbitrary points surrounding the ground target. All flight data are acquired online from the UAV's onboard navigation sensors with a rate of 5 Hz.

On the other side, the chasing network is validated after successfully approaching the desired target. The approached ground target will have different heading (orientations) at each test time. At the beginning of the chasing network, the UAV will align (by means of the yawing control) itself with the target heading then the target starts to move prompting the UAV to follow it. This leashing scenario is established to assure the ability of the UAV to align itself with arbitrary target's orientations, then how the chasing controller will steer it to follow the desired target. Figure 9 shows a five flights with aligned (almost parallel to the image x-axis) target's heading angle and the remaining three flights with different target's heading angle.

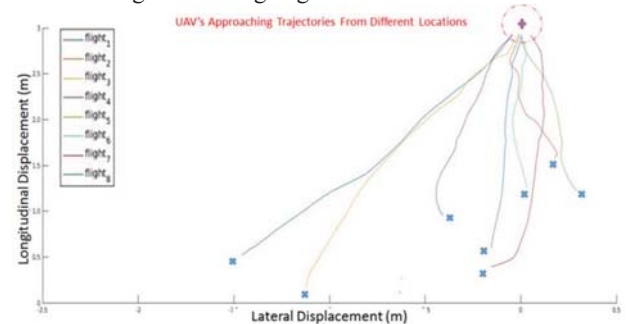


Fig. 8. The UAV is launched from some arbitrary points "x-marks" with different longitudinal and lateral displacements to approach the target "+-mark".

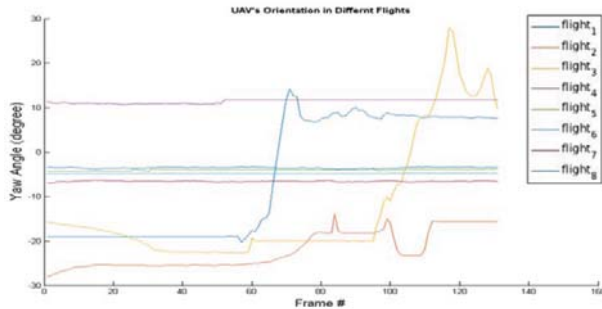


Fig. 9. Five flights with aligned (almost parallel to the image x-axis) target's heading angle and the remaining three flights with different target's heading angle.

Finally, a long-time flight (more than 4 minutes) has been examined with the proposed vision-based system to track and follow a person in outdoor environment on that video link <https://github.com/Ali-Maher/UAV-TARGET-TRACKING-BY-DETECTION-VIA-DEEP-NEURAL-NETWORKS>.

5. CONCOLOSION

A full vision-based real time deep target tracking and following system was introduced to control the UAV to follow a ground target. To the extent of our knowledge, this is the first vision-based ground target chasing proposal divides the tracking and following task into two successive phases based on two mutual deep networks. The proposed system realizes how to approach to the desired target before chasing it and resolves the ambiguity between yawing and lateral UAV's movement while the target's movement. A ground target dataset utilizing three prototypes (Jeep, Tank, and Aircraft) was built to facilitate the training of both deep networks on that link <https://github.com/Ali-Maher/Ground-Target-Dataset>.

6. ACKNOWLEDGEMENT

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